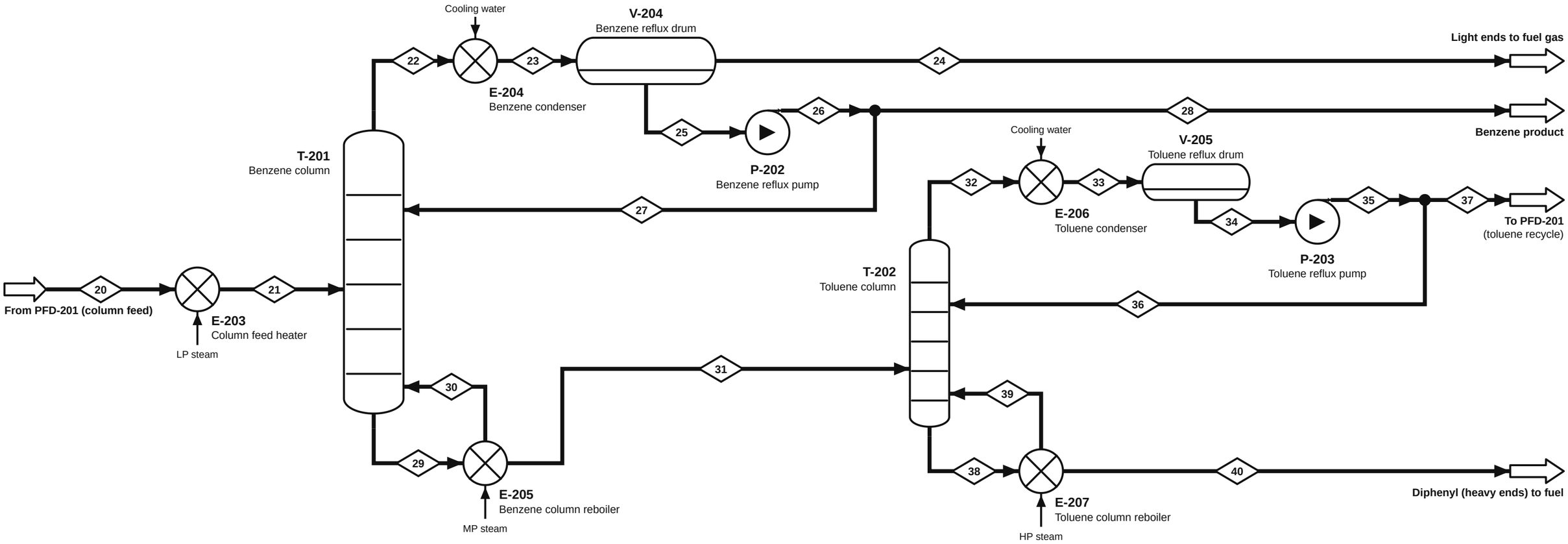
PROCESS FLOW DIAGRAM — PFD-202

Toluene hydrodealkylation to benzene | Separation section | Sheet 2 of 2



Stream table: sample values (flows from a simple component-balance model; temperatures and pressures nominal, not computed)

Stream	Phase	Temperature (°C)	Pressure (kPa)	Total (kmol/h)	Hydrogen (kmol/h)	Methane (kmol/h)	Benzene (kmol/h)	Toluene (kmol/h)	Diphenyl (kmol/h)
20	Liquid	37	280	130.2	0.1	1.2	95.9	32.5	0.5
21	Liquid	90	250	130.2	0.1	1.2	95.9	32.5	0.5
22	Vapour	100	220	210.4	0.1	1.2	208.7	0.4	0.0
23	Mixed	45	200	210.4	0.1	1.2	208.7	0.4	0.0
24	Vapour	45	200	2.3	0.1	1.2	1.0	0.0	0.0
25	Liquid	45	200	208.1	0.0	0.0	207.7	0.4	0.0
26	Liquid	45	400	208.1	0.0	0.0	207.7	0.4	0.0
27	Liquid	45	400	113.5	0.0	0.0	113.3	0.2	0.0
28	Liquid	45	400	94.6	0.0	0.0	94.4	0.2	0.0
29	Liquid	150	230	243.7	0.0	0.0	8.3	234.7	0.7
30	Vapour	152	230	210.4	0.0	0.0	7.8	202.4	0.2
31	Liquid	152	230	33.3	0.0	0.0	0.5	32.3	0.5
32	Vapour	128	160	58.5	0.0	0.0	0.9	57.6	0.0
33	Liquid	120	150	58.5	0.0	0.0	0.9	57.6	0.0
34	Liquid	120	150	58.5	0.0	0.0	0.9	57.6	0.0
35	Liquid	121	350	58.5	0.0	0.0	0.9	57.6	0.0
36	Liquid	121	350	26.0	0.0	0.0	0.4	25.6	0.0
37	Liquid	121	350	32.5	0.0	0.0	0.5	32.0	0.0
38	Liquid	270	170	59.2	0.0	0.0	0.0	52.9	6.3
39	Vapour	272	170	58.4	0.0	0.0	0.0	52.6	5.8
40	Liquid	272	170	0.8	0.0	0.0	0.0	0.3	0.5

Abbreviations

kmol/h: kilomoles per hour; kPa: kilopascals (absolute); °C: degrees Celsius
HP / MP / LP steam: high- / medium- / low-pressure steam
Mixed: vapour-liquid mixture
Stream numbers in diamonds; continuation streams keep their number on both sheets

Toluene hydrodealkylation (HDA) plant — PFD-202	
Separation section | continued on PFD-201	
Sample process flow diagram; stream values are sample values	
Originator: Claude Code | 2026-10-24	Revision 1 | Sheet 2 of 2